Davis–Kahan Perturbation Bounds for Principal Subspaces

Spectral Stability of Equity Covariance

Mathematical Foundations of Data Science
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October 2026

Abstract

Covariance estimation changes both eigenvalues and principal eigenspaces. Eigenvalue perturbations are controlled by Weyl's inequality, whereas subspace rotations depend additionally on the separation between retained and discarded eigenvalues. Beginning with a two-dimensional perturbation calculation, we express leading-subspace displacement through principal angles and derive a Davis–Kahan-type upper bound for the normalized Frobenius distance between population and perturbed top-k eigenspaces. The resulting estimate makes explicit the roles of operator-norm error and the population eigengap.

1 Covariance perturbations and spectral separation

Let $\Sigma \in \mathbb{R}^{d\times d}$ be a symmetric population covariance matrix. Its estimate is written as

$$\widehat{\Sigma} = \Sigma + E, \qquad E = E^{\mathsf{T}} \in \mathbb{R}^{d\times d}, \qquad \varepsilon = \|E\|_2. \tag{1}$$

The matrix E represents estimation error, while ε measures its magnitude in the operator norm. Suppose the eigenvalues of Σ are ordered as

$$\lambda_1 \geq \cdots \geq \lambda_d,$$

and fix $1 \leq k < d$. Let $\mathcal{U}$ be the invariant subspace associated with the largest k eigenvalues of Σ, and let $\widehat{\mathcal{U}}$ be its estimated analogue associated with the largest k eigenvalues of $\widehat{\Sigma}$.

Two quantities govern stability: the perturbation size ε and the *population eigengap*

$$\delta = \lambda_k - \lambda_{k+1} > 0. \tag{2}$$

A larger gap gives greater spectral separation between the leading and discarded eigenvectors. We seek a quantitative bound on the angle between $\mathcal{U}$ and $\widehat{\mathcal{U}}$. We will use Weyl's eigenvalue perturbation inequality,

$$|\widehat{\lambda}_j - \lambda_j| \leq \|E\|_2 = \varepsilon, \qquad j = 1, \ldots, d, \tag{3}$$

where $\widehat{\lambda}_1 \geq \cdots \geq \widehat{\lambda}_d$ denote the eigenvalues of $\widehat{\Sigma}$.

2 A two-dimensional motivating calculation

First consider

$$\Sigma = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}, \quad \lambda_1 > \lambda_2 \geq 0, \qquad E = \begin{pmatrix} a & b \\ b & c \end{pmatrix}. \tag{4}$$

Write the leading unit eigenvector of $\widehat{\Sigma}$ as

$$\widehat{v}_1 = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2},$$

after selecting its sign so that $\cos\theta \geq 0$. The angle between the *lines* spanned by e_1 and $\widehat{v}_1$ is $|\theta|$, and the sine of this angle is $|\sin\theta|$.

Proposition 2.1 (Rotation of a leading eigenvector)**.** *If $\varepsilon < \lambda_1 - \lambda_2$, then*

$$|\sin\theta| \le \frac{\varepsilon}{\lambda_1 - \lambda_2 - \varepsilon}. \tag{5}$$

In particular, if $\varepsilon \le (\lambda_1 - \lambda_2)/2$, then

$$|\sin\theta| \le \frac{2\varepsilon}{\lambda_1 - \lambda_2}. \tag{6}$$

Proof. The eigenvalue equation is

$$(\Sigma + E)\widehat{v}_1 = \widehat{\lambda}_1\widehat{v}_1, \qquad (\widehat{\lambda}_1 I - \Sigma)\widehat{v}_1 = E\widehat{v}_1.$$

Taking the second coordinate gives

$$(\widehat{\lambda}_1 - \lambda_2)\sin\theta = b\cos\theta + c\sin\theta = e_2^{\mathsf{T}} E\widehat{v}_1. \tag{7}$$

Since $\|e_2\|_2 = \|\widehat{v}_1\|_2 = 1$, Cauchy–Schwarz and the definition of the operator norm imply

$$|e_2^{\mathsf{T}} E\widehat{v}_1| \le \|E\widehat{v}_1\|_2 \le \varepsilon.$$

By Weyl's inequality,

$$\widehat{\lambda}_1 - \lambda_2 \ge \lambda_1 - \varepsilon - \lambda_2 = (\lambda_1 - \lambda_2) - \varepsilon > 0.$$

Dividing (7) by this positive quantity proves (5). Under $\varepsilon \le (\lambda_1 - \lambda_2)/2$, the denominator is at least $(\lambda_1 - \lambda_2)/2$, giving (6). □

The estimate has a geometric meaning: for one-dimensional subspaces, the operator-norm difference of their orthogonal projectors equals the sine of their principal angle. Thus $|\sin\theta|$ measures the rotation of the leading eigendirection without depending on eigenvector sign.

3 Geometry of the top-k subspace

For the general case, let

$$V_k = [v_1 \ \cdots \ v_k] \in \mathbb{R}^{d\times k}, \qquad V_\perp = [v_{k+1} \ \cdots \ v_d] \in \mathbb{R}^{d\times(d-k)} \tag{8}$$

contain orthonormal eigenvectors of Σ. Then

$$V_k^{\mathsf{T}} V_k = I_k, \qquad V_\perp^{\mathsf{T}} V_\perp = I_{d-k}, \qquad V_k V_k^{\mathsf{T}} + V_\perp V_\perp^{\mathsf{T}} = I_d.$$

Let $\widehat{V}_k \in \mathbb{R}^{d\times k}$ contain orthonormal leading eigenvectors of $\widehat{\Sigma}$. Define the *discarded-space component* of the estimated eigenvectors by

$$W = V_\perp^{\mathsf{T}} \widehat{V}_k \in \mathbb{R}^{(d-k)\times k}. \tag{9}$$

Its (i, j) entry records the component of $\widehat{v}_j$ along v_{k+i}. Consequently, $W = 0$ exactly when the estimated leading subspace lies inside the population leading subspace; since they have the same dimension, this means the two subspaces agree.

Lemma 3.1 (Discarded components and principal angles)**.** *If $\theta_1, \ldots, \theta_k \in [0, \pi/2]$ are the principal angles between $\operatorname{im} V_k$ and $\operatorname{im} \widehat{V}_k$, then*

$$\boxed{\|W\|_F^2 = \sum_{j=1}^{k} \sin^2\theta_j.} \tag{10}$$

Proof. Set $C = V_k^{\mathsf{T}} \widehat{V}_k \in \mathbb{R}^{k\times k}$. From the orthogonal decomposition of the identity,

$$\begin{aligned} W^{\mathsf{T}} W + C^{\mathsf{T}} C &= \widehat{V}_k^{\mathsf{T}} (V_\perp V_\perp^{\mathsf{T}} + V_k V_k^{\mathsf{T}}) \widehat{V}_k \\ &= \widehat{V}_k^{\mathsf{T}} \widehat{V}_k = I_k. \end{aligned}$$

Taking traces yields

$$\|W\|_F^2 = k - \|C\|_F^2. \tag{11}$$

The singular values of C are $\cos\theta_1, \ldots, \cos\theta_k$. Hence

$$\|C\|_F^2 = \sum_{j=1}^{k} \cos^2\theta_j, \qquad \|W\|_F^2 = \sum_{j=1}^{k} (1 - \cos^2\theta_j) = \sum_{j=1}^{k} \sin^2\theta_j.$$

□

This identity makes the connection to the normalized projector distance immediate. With $P = V_k V_k^{\mathsf{T}}$ and $\widehat{P} = \widehat{V}_k \widehat{V}_k^{\mathsf{T}}$, define

$$d_k(\mathcal{U}, \widehat{\mathcal{U}}) := \frac{\|P - \widehat{P}\|_F}{\sqrt{2k}} = \left(\frac{1}{k} \sum_{j=1}^{k} \sin^2\theta_j \right)^{1/2} = \frac{\|W\|_F}{\sqrt{k}}. \tag{12}$$

4 A Davis–Kahan-type bound

Put

$$\Lambda_\perp = \operatorname{diag}(\lambda_{k+1}, \ldots, \lambda_d) \in \mathbb{R}^{(d-k)\times(d-k)}, \qquad \widehat{\Lambda}_k = \operatorname{diag}(\widehat{\lambda}_1, \ldots, \widehat{\lambda}_k) \in \mathbb{R}^{k\times k}. \tag{13}$$

We now relate W to the perturbation E through the eigenvalue equations.

Theorem 4.1 (Top-k spectral-subspace stability)**.** *Let Σ and $\widehat{\Sigma} = \Sigma + E$ be symmetric $d \times d$ matrices, with eigenvalues ordered in decreasing order. If the population eigengap $\delta = \lambda_k - \lambda_{k+1}$ is positive and $\varepsilon = \|E\|_2 < \delta$, then*

$$\boxed{d_k(\mathcal{U}, \widehat{\mathcal{U}}) \le \frac{\varepsilon}{\delta - \varepsilon}.} \tag{14}$$

In particular, under the sufficient assumption $\varepsilon \le \delta/2$,

$$\boxed{d_k(\mathcal{U}, \widehat{\mathcal{U}}) \le \frac{2\|E\|_2}{\delta}.} \tag{15}$$

Proof. The leading estimated eigenvectors satisfy

$$\widehat{\Sigma}\widehat{V}_k = \widehat{V}_k \widehat{\Lambda}_k.$$

Substitute $\widehat{\Sigma} = \Sigma + E$ and multiply on the left by $V_\perp^{\mathsf{T}}$:

$$V_\perp^{\mathsf{T}} \widehat{V}_k \widehat{\Lambda}_k - V_\perp^{\mathsf{T}} \Sigma \widehat{V}_k = V_\perp^{\mathsf{T}} E \widehat{V}_k. \tag{16}$$

Because $\Sigma V_\perp = V_\perp \Lambda_\perp$ and Σ is symmetric, $V_\perp^{\mathsf{T}} \Sigma = \Lambda_\perp V_\perp^{\mathsf{T}}$. Using $W = V_\perp^{\mathsf{T}} \widehat{V}_k$, we obtain

$$\boxed{W\widehat{\Lambda}_k - \Lambda_\perp W = V_\perp^{\mathsf{T}} E \widehat{V}_k.} \tag{17}$$

The two diagonal matrices make this equation entrywise:

$$(\widehat{\lambda}_j - \lambda_{k+i}) W_{ij} = (V_\perp^{\mathsf{T}} E \widehat{V}_k)_{ij}, \quad 1 \le i \le d-k, \quad 1 \le j \le k. \tag{18}$$

By Weyl's inequality and the ordering of eigenvalues,

$$\begin{aligned} \widehat{\lambda}_j - \lambda_{k+i} &\ge \lambda_j - \varepsilon - \lambda_{k+i} \\ &\ge \lambda_k - \lambda_{k+1} - \varepsilon = \delta - \varepsilon > 0. \end{aligned}$$

Thus the entries of W obey

$$|W_{ij}| \le \frac{|(V_\perp^{\mathsf{T}} E \widehat{V}_k)_{ij}|}{\delta - \varepsilon}.$$

Squaring and summing over both indices gives

$$\|W\|_F^2 \le \frac{\|V_\perp^\mathsf{T} E \widehat{V}_k\|_F^2}{(\delta-\varepsilon)^2}. \tag{19}$$

Since $V_\perp V_\perp^\mathsf{T}$ is an orthogonal projector, it cannot increase Euclidean vector norms. Therefore, writing $\widehat{v}_j$ for the columns of $\widehat{V}_k$,

$$\begin{aligned}\|V_\perp^\mathsf{T} E \widehat{V}_k\|_F^2 &= \sum_{j=1}^{k} \|V_\perp^\mathsf{T} E \widehat{v}_j\|_2^2 \\ &\le \sum_{j=1}^{k} \|E\widehat{v}_j\|_2^2 \le \sum_{j=1}^{k} \varepsilon^2 = k\varepsilon^2.\end{aligned}$$

It follows that

$$\|W\|_F^2 \le \frac{k\varepsilon^2}{(\delta-\varepsilon)^2}. \tag{20}$$

Finally, the geometric identity (12) yields

$$d_k(\mathcal{U},\widehat{\mathcal{U}})^2 = \frac{\|W\|_F^2}{k} \le \frac{\varepsilon^2}{(\delta-\varepsilon)^2}.$$

Taking square roots establishes (14). If $\varepsilon \le \delta/2$, then $\delta - \varepsilon \ge \delta/2$, so (15) follows. □

5 Interpretation for covariance estimation

The estimate separates two drivers of subspace displacement: the covariance estimation error $\|E\|_2$ and the spectral gap $\delta = \lambda_k - \lambda_{k+1}$. Under sufficiently small perturbations relative to this gap, the normalized distance between the leading estimated and population eigenspaces is controlled by their ratio. A small gap makes the upper bound weaker; by itself, it does not establish that a particular observed subspace must rotate substantially.

For rolling equity-covariance estimates, this bound provides a theoretical benchmark against which measured principal-subspace drift can be interpreted. Changes in an estimated subspace may reflect sampling error as well as changes in the underlying covariance structure; the theorem alone does not distinguish them.